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(54) **DAMAGE ASSESSMENT**

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(57) **ABSTRACT**

Methods of estimating tribological damage described herein include examples where varying power is applied between surfaces engaged in frictional contact. Calculations evaluate power consumed at the relevant frictional contact and temperature values may be gathered to supplement the calculated power. Instantaneous and cumulative assessments of damage are calculated based on that information. Measurements or calculations of electrical power may be used as part of the damage assessment.